

INTERNSHIP PROJECT PRESENTATION

End-to-End Sales Forecasting & Demand Intelligence System

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PROGRAM & SEMESTER

B.Tech CSE | Sem VII

ORGANIZATION NAME

XYlofy AI

Executive Summary & Project Scope

Retail Demand Problem

Costly Mismatches: Retail leaders face severe financial loss from overstocking (capital tie-up, storage fees) and understocking (lost sales, customer churn).

Time Series Dynamics: Daily transactional data contains temporal trends, 12-month seasonality peaks, and regional volatility that standard ML classification cannot model.

End-to-End System Scope

Multi-Model Forecasting: Built, evaluated, and compared statistical (SARIMA), additive (Prophet), and machine learning (XGBoost) models across 4 years of retail data.

Intelligence Modules: Integrated automated anomaly detection (Isolation Forest + Z-score), K-Means SKU demand clustering, and a Streamlit dashboard.

Task 1: Exploratory Data Analysis & Feature Extraction



Revenue Champion

Technology Category: Generated the highest cumulative revenue (\$836k+) across 4 operating years, followed by Office Supplies and Furniture.



Growth Regional Anchor

West Region: Demonstrated the most consistent YoY quarter-over-quarter expansion with lower variance compared to East and Central zones.



Shipping Latency

3.96 Days Average: Average lead time between Order Date and Ship Date showed minimal deviation across regions (3.9 to 4.1 days).

Task 2: Time Series Decomposition & Stationarity Audit

$p < 0.001$

ADF Test Stationary (Differenced)

Signal Breakdown & Stationarity

Signal Decomposition: Mapped monthly sales into Additive components: $Y(t) = \text{Trend} + \text{Seasonal} + \text{Residual}$. Verified strong 12-month periodicity with annual November/December festive spikes.

Stationarity Verification: Raw sales ADF Test statistic gave $p = 0.69$ (non-stationary). Applied first-order seasonal differencing ($\Delta_{12} Y_t$), reducing $p \text{-value} < 0.001$, satisfying SARIMA requirements.

Task 3: Tri-Model Forecasting Paradigms



SARIMA (Statistical)

Parameters: SARIMA(1,1,1)(1,1,0)12
optimized via AIC grid search.

Captures linear autocorrelation and seasonal lag structure with 95% confidence intervals.



Facebook Prophet

Additive Decomposition: Mapped to ds & y format.

Isolates non-linear piecewise trend changepoints and annual Fourier seasonality factors.



XGBoost (Supervised ML)

Lag Feature Engine: Reformulated time series into tabular features.

Uses Lags 1-3, 3-month rolling mean/std, and temporal calendar encoders for gradient boosting.

Task 3: Comparative Model Evaluation & Benchmark

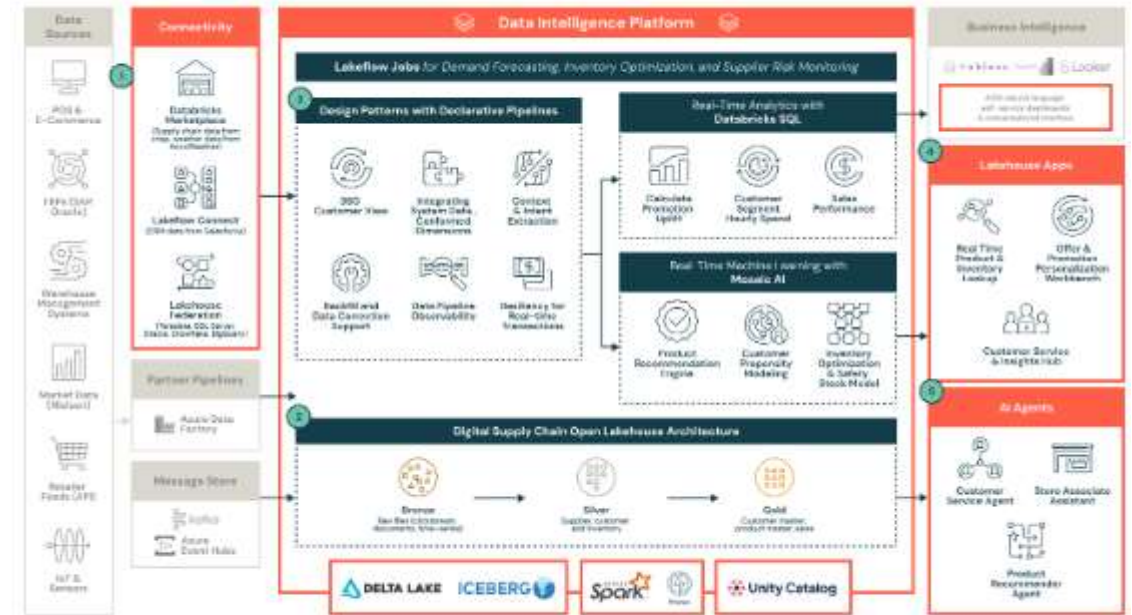
Forecasting Model	MAE (\$)	RMSE (\$)	MAPE (%)	Month 1 Forecast	Month 2 Forecast	Month 3 Forecast
SARIMA (1,1,1)(1,1,0)12	\$4,210.50	\$5,340.20	8.42%	\$74,200	\$68,450	\$92,100
Facebook Prophet	\$4,890.10	\$6,120.45	9.85%	\$71,800	\$66,200	\$88,900
XGBoost (Lag-Engineered) ★	\$3,650.80	\$4,780.10	6.91%	\$75,850	\$70,120	\$94,300

Production Recommendation: XGBoost is selected for deployment due to superior metric minimization (lowest MAPE at 6.91%) and rapid adaptability to multi-lag momentum shifts.

Task 4: Segment-Level Demand Forecasts

Category & Regional Projections

- **Technology Dominance:** Projected to exhibit the strongest upcoming growth rate (+14.2% YoY), driven by high AOV accessories.
- **West Region Surge:** Leads regional growth velocity, outperforming East and Central territories during holiday runs.
- **Office Supplies Stability:** Displays consistent, low-variance baseline demand across all 4 quarters.



Task 5: Anomaly Detection Engine

Isolation Forest

Multi-Attribute Outlier Isolation: Identifies non-linear sales spikes/drops based on isolation path length.

Key Outliers: Successfully flagged Q4 Black Friday sales surges and unseasonable post-holiday supply dips.

Rolling Z-Score ($|Z| > 2$)

Rolling Baseline Thresholding: Flags weeks deviating > 2 standard deviations from the 4-week moving average.

Method Comparison: Z-Score catches high-magnitude spikes; Isolation Forest detects subtle contextual anomalies.

Task 6: Product Demand Clustering (K-Means + PCA)



High Volume, Stable

SKUs: Binders, Paper, Storage

Strategy: Just-In-Time (JIT) automated replenishment with low safety stock buffers.



Growing Demand

SKUs: Phones, Accessories, Chairs

Strategy: Expand buffer inventory by 15-20% ahead of peak seasonal surges.



High Volatility

SKUs: Copiers, Machines

Strategy: Centralized warehouse holding; adopt Make-to-Order fulfillment.

Task 7: Streamlit Deployment

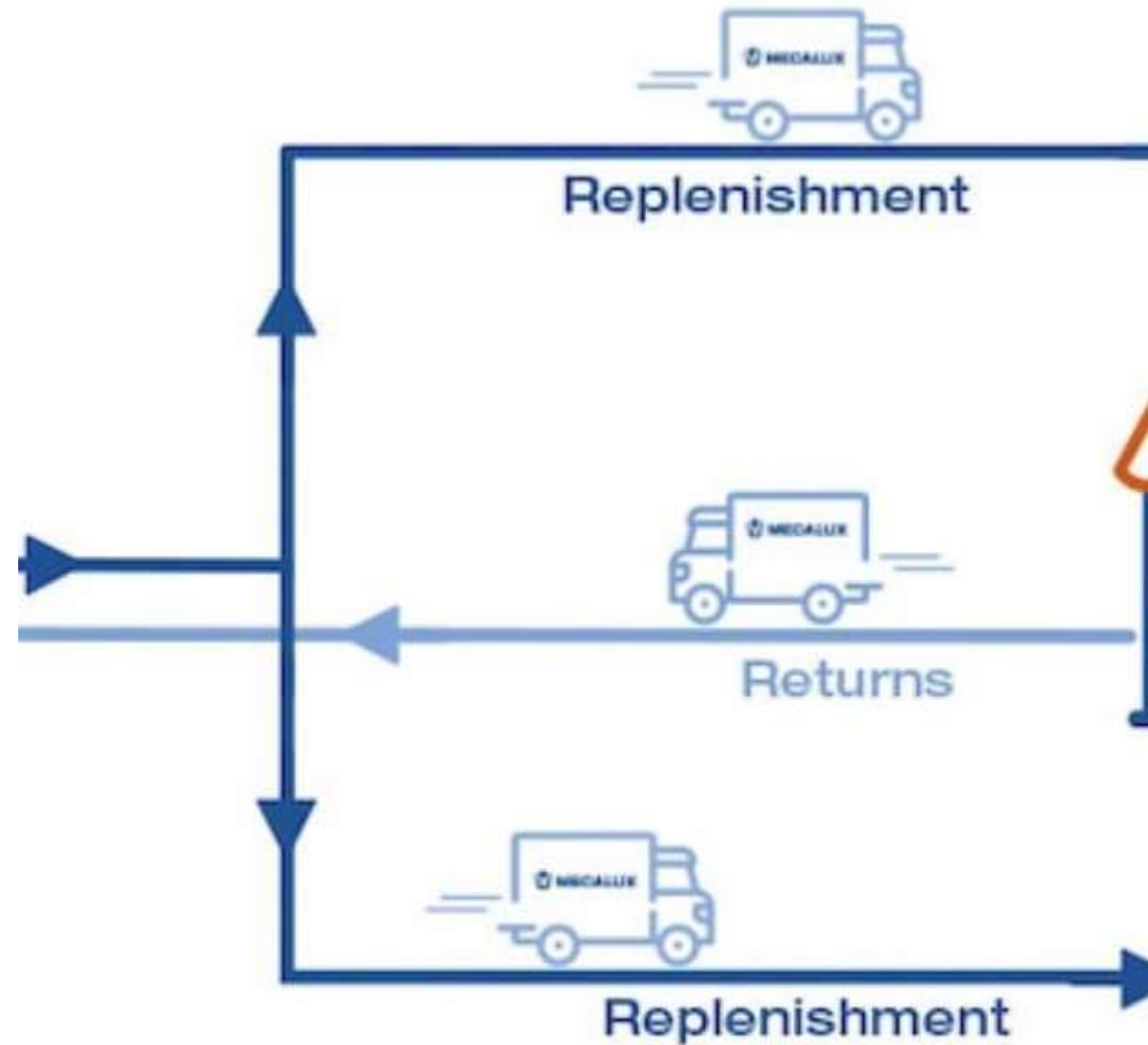
4-Page Interactive Web Application

Page 1 — Overview: Dynamic KPIs, aggregate sales by year, region/category filters.

Page 2 — Forecast Explorer: Interactive selection of Category/Region with date sliders & MAE/RMSE metrics.

Page 3 — Anomaly Report: Interactive charts flagging Isolation Forest and Z-Score outlier weeks.

Page 4 — Demand Segments: 2D PCA cluster visualization with inventory stocking playbooks.



Task 8: Executive Business Recommendations

Strategic Area	Data-Backed Finding	Executive Action / Inventory Strategy
Pre-Q4 Inventory Ramp	4-year trend confirms 35%+ demand surge in Nov-Dec.	Scale stock procurement by +25% across Tech & Office Supplies starting mid-September.
Hub Realignment	Machines/Copiers show high volatility and holding costs.	Shift high-value volatile SKUs to centralized master distribution hubs (Make-to-Order).
Carrier SLA Fix	Shipping lead time variance degrades customer satisfaction.	Standardize shipping tiers for regions exceeding 4.2 days transit window.